TEST PROCEDURE

FOR

POWER DISTRIBUTION UNIT (PDU)

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Prepared by
HBL Engineering Ltd, Hyderabad
(Formerly known as HBL Power Systems Ltd)

DOCUMENT DATA SHEET

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Abstract

This document details Test Procedure for IPS PDU of KAVACH (IRATP)

DOCUMENT CONTROL SHEET

#	Name	Organization	Function	Level
1	Mr. JVVSNV Prasad	HBL	Engineering	Approve
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CHANGE HISTORY

#	Name of the Document	Date	Reason for changes	Version No.
1.	Test Procedure for PDU	18/03/2026	Initial document	1.0

TABLE OF CONTENTS

1. PURPOSE:

The purpose of this document is to give comprehensive guidelines and set of procedures to be adopted for carrying out the test procedure for IPS PDU.

2. SCOPE:

The instructions are prepared to cover all the activities involved in IPS PDU Test Procedure.

3. RESPONSIBILITY AND AUTHORITY:

The QC in charge shall be responsible for implementing and maintaining the test instructions.

4. TOOLS & EQUIPMENT REQUIRED:

- Multimeter
- 110 DC Source
- PDU (Station PDU, IPS PDU, FAT SETUP PDU, LAB MODEL PDU)

5. DOCUMENTS REQUIRED:

BOM	:	6000062651_BOM_IPSIP SOURCE_POWER_ DISTRIBUTION_MODULE 6000051059_MLBOM_TCAS_POWER_DISTRIBUTION_MODULE 6000064453_HARNS_ASSY_KAVACH_FATSETUP_PDU_516490709 6000064489_KAVACH_LAB_MODEL_POWER_DISTRIBUTION_UNIT
Harness	:	6000062977 HARNS_ASSY_IP_SOURCE_IPS_PDU_516490671 6000051277_HARNS_ASSY_PWR_DISTRIBUT_UNIT_516490451 6000064453_HARNS_ASSY_KAVACH_FATSETUP_PDU_516490709 6000064488_HARNS_ASSY_LAB_MODEL_PDU_516490720

6. PROCEDURE

6.1 IPS PDU

6.2.1 Give the input 110VDC to the IPS PDU at MCB 1 terminals from 110VDC Source.

6.2.2 Measure the 110VDC at the output of MCB 1 by switching the MCB 1 to ON Position.

6.2.3 Now switch ON the MCB 2 and check that 110VDC is available at RTU 1 Terminal Blocks with the help of Multimeter.

6.2.4 Similarly check the voltages at RTU 2, STN PDU 1 & STN PDU 2, RIU 1 & RIU 2 Terminal Blocks by switching ON the MCB 3, MCB 4, MCB 5, MCB 6, MCB 7 respectively.

6.2 Station PDU

6.3.1 Take two 110 VDC Source and give input 110VDC from Source 1 to MCB 1 and Source 2 to MCB 2.

6.3.2 Check the 110VDC at the outputs of MCB 1 & MCB 2 by switching ON the MCBs and Sources with the help of Multimeter.

6.3.3 Now, Switch ON only MCB 1 and check the respective voltages as per table with the help of Multimeter.

When MCB 1 is ON		
MCB	**Output TBs**	**Expected Voltage**
MCB 3	N/W Rack 1 Terminal Blocks	110 VDC
MCB 5	DC-DC Conv 1 Terminal Blocks	110 VDC
MCB 7	RIU 1 Terminal Blocks	110 VDC
MCB 9	DPS 1 Terminal Blocks	110 VDC

6.3.4 Now, Switch ON only MCB 2 and check the respective voltages as per table with the help of Multimeter.

When MCB 2 is ON		
MCB	**Output TBs**	**Expected Voltage**
MCB 4	N/W Rack 2 Terminal Blocks	110 VDC
MCB 6	DC-DC Conv 2 Terminal Blocks	110 VDC
MCB 8	RIU 2 Terminal Blocks	110 VDC
MCB 10	DPS 2 Terminal Blocks	110 VDC

6.3 FAT SETUP PDU

6.4.1 Take two 110 VDC Source and give input 110VDC from Source 1 to MCB 1 and Source 2 to MCB 2.

6.4.2 Check the 110VDC at the outputs of MCB 1 & MCB 2 by switching ON the MCBs and Sources with the help of Multimeter.

6.4.3 Now, Switch ON only MCB 1 and check the respective voltages as per table with the help of Multimeter.

When MCB 1 is ON		
MCB	**Output TBs**	**Expected Voltage**
MCB 3	RTU 1 Terminal Blocks	110 VDC
MCB 5	LTCAS 1 Terminal Blocks	110 VDC
MCB 7	DPS 1 Terminal Blocks	110 VDC

6.4.4 Now, Switch ON only MCB 2 and check the respective voltages as per table with the help of Multimeter.

When MCB 2 is ON		
MCB	**Output TBs**	**Expected Voltage**
MCB 4	RTU 2 Terminal Blocks	110 VDC
MCB 6	LTCAS 2 Terminal Blocks	110 VDC
MCB 8	DPS 2 Terminal Blocks	110 VDC

6.4 LAB MODEL PDU

6.4.1 Take two 110 VDC Source and give input 110VDC from Source 1 to MCB 1 and Source 2 to MCB 2.

6.4.2 Check the 110VDC at the outputs of MCB 1 & MCB 2 by switching ON the MCBs and Sources with the help of Multimeter.

6.4.3 Now, Switch ON only MCB 1 and check the respective voltages as per table with the help of Multimeter.

When MCB 1 is ON		
MCB	**Output TBs**	**Expected Voltage**
MCB 3	RTU 1 Terminal Blocks	110 VDC
MCB 5	LTCAS EMI-FILTER Terminal Blocks	110 VDC
MCB 7	LTCAS CAB I/P Terminal Blocks	110 VDC
MCB 9	DPS 2 Terminal Blocks	110 VDC

6.4.4 Now, Switch ON only MCB 2 and check the respective voltages as per table with the help of Multimeter.

When MCB 2 is ON		
MCB	**Output TBs**	**Expected Voltage**
MCB 4	RTU 2 Terminal Blocks	110 VDC
MCB 6	LTCAS RIB Terminal Blocks	110 VDC
MCB 8	DPS 1 Terminal Blocks	110 VDC